

Checklist

What you should know

After studying this subtopic, you should be able to:

- Recognise nucleophiles in chemical reactions.
- Deduce equations with descriptions and explanations of the movement of electron pairs in nucleophilic substitution reactions.
- Explain, with equations, the formation of ions by heterolytic fission.
- Recognise electrophiles in chemical reactions.
- Deduce equations for the reactions of alkenes with water, halogens and hydrogen halides.

Higher level (HL)

- Apply Lewis acid—base theory to inorganic and organic chemistry to identify the role of the reacting species.
- Explain the relationship between Bronsted—Lowry acids and bases and Lewis acids and bases.
- Draw and interpret Lewis formulas of reactants and products to show coordination bond formation in Lewis acid—base reactions.
- Deduce the charge on a complex ion, given the formula of the ion and ligands present.
- Describe and explain the mechanisms of the reactions of primary and tertiary halogenoalkanes with nucleophiles.
- Explain the stereospecific nature of S_N2 .
- Predict and explain the relative rates of the substitution reactions for different halogenoalkanes (RCI , RBr , RI).
- Describe and explain the mechanisms of the reactions between symmetrical alkenes and halogens, water and hydrogen halides.
- Predict and explain the major product of a reaction between an unsymmetrical alkene and a hydrogen halide or water.
- Describe and explain the mechanism of the reaction between benzene and a charged electrophile, E^+ .

- Describe the features of benzene (C_6H_6) that make it not prone to undergo addition reactions, despite being highly unsaturated.